

Complex numbers



1 A complex number

2 $0 + bi$ or bi

3 a $\sqrt{-1}$

b -1

4 a 0

b $-\sqrt{10}$

c Nonreal complex/pure imaginary

5 a $\sqrt{5}i$

b $\sqrt{14}$

Operations with complex numbers

6 $9 + 2i$

7 No, $-4i$

8 No, $\frac{6 \pm 36i}{2} = 3 \pm 18i$

9 a $-10i$

b $3i\sqrt{2}$

c $9 - 8i$

d $2 + 5i$

10 -8

11 a -19

b 5

c $\sqrt{5}$

d $\frac{1}{3}$

12

a $10 + 9i$

b $3 + 9i$



c $-3 + 5i$

d $6 - 11i$

e $-3.3 + 2.6i$

f $-11 + 16i$

g $-11 - 5i$

h $12 - 6i$

i $\frac{7}{6} - \frac{i}{20}$

13

a $-18 + 30i$

b $14 + 21i$

c $-10 + 8\sqrt{10}i$

d $96 - 16i$

e 65

f -29

g $-21 - 20i$

h $-\frac{1}{4} + \frac{3}{4}i$

i $16i + 12$

j $-14i - 48$

14

a

i -1

ii 1

iii -1

iv 1

b

i Yes

ii No

iii Yes

iv No

15

a $-2i$

b $-v$

c -4

d 10

e $-27i$

f 16

g $-13 - 7i$

h $-7 - 2i$

Complex conjugates

16 Complex conjugates

17 A real number

18

a 20

b 39

c 26

d 25

e 17

f 3

g 21

h 5

19

a $2 - 3i$

b $-5i + 2$

c $\frac{5i}{6}$

d $2 + 5i$

e $5 + i$

f $2 - i$

g $3 + 2i$

h $-\frac{5}{13} - \frac{12i}{13}$

i $-\frac{2}{17} - \frac{9i}{17}$

j $\frac{2}{5} + \frac{6i}{5}$

20

a 10



b 31

21

a $\frac{67 - 47i}{34}$

b $-7 - 10i$

22

$\frac{16}{5} - 5i$

Application

23

$33 + 50i$